Predicting Bitcoin's Hourly Price Changes: The Direction Is Noise, the Size Is Not

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***Abstract*—We study how to forecast the 24 hourly price changes of Bitcoin's next day from the preceding hourly data. On 59,331 consecutive hourly Bitstamp candles (January 2020 – October 2026), we build scale-free features, split the data into chronological train, validation and test blocks with a purge between them, make every choice without the test block, and compare regularised linear models, nearest neighbours, support vector machines, tree ensembles, neural networks (MLP, LSTM) and a pretrained time-series foundation model (Timestral, zero-shot) against simple rules, with block-bootstrap confidence intervals. No model forecasts the hourly returns better than "no change": at the hourly scale Bitcoin behaves like a random walk, and only the next hour carries a small reversal signal (AUC 0.538). The *size* of the moves, however, is clearly predictable. Gradient Boosting refitted monthly explains 14.8% of the variance of each hourly absolute return (2.6 times a volatility rule), an LSTM fine-tuned monthly 15.6%, zero-shot Timestral 15.7%, and the average of Timestral and Gradient Boosting 16.4%. Conformally calibrated 80% ranges cover 80% of the test hours, and next-day volatility is predicted with $R^2 \approx 0.51$. In a $10,000 trading simulation these risk forecasts earn no more than a naive volatility estimate, but make trading safer: fewer liquidations at the same leverage and a lower chance of a large drawdown. Keeping models up to date mattered more than any extra feature or data source.**

***Index Terms*—Bitcoin, time-series forecasting, volatility, gradient boosting, conformal prediction, LSTM, foundation models**

I. Introduction

The assigned task is to predict, for each of the 24 hours of the next day, how much the price of Bitcoin will change, $r_h = \mathrm{Close}_h / \mathrm{Close}_{h-1} - 1$, from the hours before. We forecast changes rather than prices because the price level is not stationary: it rose from about $4,000 (March 2020) to over $126,000, and 72% of the test hours trade above the highest training price, so a model of the level would have to extrapolate.

Short-horizon forecasts matter for risk management, position sizing and the timing of trades, and Bitcoin is a demanding case: it trades 24/7, its hourly returns are heavy-tailed, and its behaviour changes between bull and bear markets and after events such as the launch of the US spot ETFs in January 2024. Two classical results frame the problem. In an efficient market, past prices should not predict the direction of future returns [1]; the size of the moves, on the other hand, clusters in time and is predictable [2], [3]. Financial time series also make it easy to fool oneself: overlapping windows and random splits leak the future into the training data [4].

Our contributions are: (i) a leakage-free evaluation in which every choice is made without the test block and every model is compared with a simple rule; (ii) evidence that hourly returns are not predictable beyond a tiny one-hour reversal, while their size is; (iii) calibrated 80% ranges for every hour of the next day; and (iv) a comparison of classical models, neural networks and a zero-shot foundation model on identical rows.

II. Methodology

A. Dataset and Source

The data is the Kaggle "Bitcoin Historical Data" set [5] of 1-minute Bitstamp BTC/USD candles, pinned to one version so that every run sees the same file, and aggregated to hourly candles (`Open`, `High`, `Low`, `Close`, `Volume`). The models use the 59,331 consecutive hours from 1 January 2020 to 8 October 2026. Auxiliary sources were tested as extra inputs: hourly BTC and ETH candles with order flow from Binance [6], about 40 macro and market series from FRED [7] and Yahoo Finance, and 281 US release times (CPI, payrolls, FOMC) from the BLS and Federal Reserve calendars.

B. Preprocessing

Minutes are aggregated to hours without dropping missing rows, so the hourly grid stays continuous. The minute data hides gaps: a minute without trades appears as a flat, zero-volume candle at the previous close. Such minutes make up 95% of 2012 and 25–40% of 2013–2016, against 1–4% per year since 2018, so the base series starts in 2018; the models use it from 2020 so that the training block is closer to the current market. Outage hours are flagged, not removed; Bitstamp and Binance hourly returns correlate at 0.993, so the data is clean. Scaling is fitted on training data only, inside each model pipeline; the LSTM normalises each input window by itself, and Timestral normalises its own inputs.

C. Exploratory Data Analysis

Analyses that could steer a model are fitted on the training block and checked on validation, never on test. Hourly returns

are heavy-tailed (excess kurtosis 55). Fig. 1 shows the central fact of the project: the autocorrelation of returns is tiny (about -0.03 at lags 1, 2 and 24, and with a band that allows for volatility clustering only the lag-24 reversal stays significant; an AR(24) model has validation $R^2 = -0.002$), while the absolute returns are strongly autocorrelated (0.30 at lag 1, still 0.10 a week later; an AR(24) on $|r|$ explains 10% of the validation variance). The direction is almost noise; the size is not.

The size also follows the clock, and this rhythm drifts (Fig. 2): the 14 UTC peak (US session) rises from 1.28 times the average hour before the spot ETFs to 1.67 after them, a change that starts in 2022 and jumps in 2024; weekend hours went from 0.8 to about 0.6 times weekday hours. Volatility regimes persist (the next day moves 0.32% per hour after calm weeks, 0.71% after wild ones), but tell how much Bitcoin will move, not which way. ETH moves with Bitcoin in the same hour (correlation 0.84) but does not lead it. Macro variables (Spearman correlations, false-discovery-rate correction [8]) move with Bitcoin (S&P 500: 0.33 daily) but none leads it. US releases do move it: the release hour moves 2.2–2.5 times the same hour on nearby days without a release (FOMC up to 3.6 times; Wilcoxon $p < 0.001$ on train and validation). From these findings, candidate features (volatility ratios, minutes without trades, release-hour flags, a 90-day hour profile) were built, checked for look-ahead and screened on train and validation.

D. Prediction Targets and Features

A forecast is made at every hour (the *origin*) that has 30 days of history before it and 24 target hours after it. Besides the assigned target, the 24 hourly returns, we ask the questions the data can answer: (a) the direction of the next hour; (b) the size $|r|$ of each of the next 24 hours; (c) an 80% prediction range for each of these hours; and (d) the next day's return, direction and volatility. All features are scale-free, so they mean the same in 2020 and 2026: lagged returns, realised volatilities over 24, 72 and 168 hours, high–low ranges, volume ratios and the calendar (the *base* set). In the *long format* each origin gives 24 rows, one per predicted hour, with per-hour columns (the average size of that hour, its size one and seven days earlier). One model then learns the common structure ("volatile weeks make every hour volatile") from 24 times more rows, and the hour-specific part from the per-hour columns.

E. Models

Classical: Ridge [9], Lasso [10], ElasticNet [11], k-nearest neighbours, SVR/SVC, Random Forest [12], Extra Trees [13], Gradient Boosting (GB) [14] and averages of them. Ranges come from GB with the quantile loss [15] for the 10% and 90% quantiles, recalibrated by *rolling conformal calibration* [16], [17]: the half-width is multiplied by the factor that would have given 80% coverage over the last 30 days of known outcomes; the *Mondrian* variant does this separately for release hours. Both volatility targets are also refitted every month on all data known by then: with weights that halve every 12 months for the hourly size, with an expanding window for next-day volatility (each chosen on validation). *Neural:* MLPs (scikit-learn, PyTorch) and LSTMs [18] reading a 168-hour window of ten window-normalised channels, trained with MSE or Huber loss. The best LSTM also receives the clock and release flags of each predicted hour, reads the hidden states of the whole week weighted by traded volume (an attention mask), and is fine-tuned every month. *Foundation model:* Timestral [19] is pretrained by Mistral AI on mostly synthetic series, reads a long context in patches of 32 hours and returns 21 quantiles of each of the next 24 hours. It is used zero-shot: its weights never see our data; we only choose, on validation, which series it reads (r or $|r|$), the context (30–180 days), the covariates and the model size. Its expected $|r|$ is the mean of the forecast distribution, since R^2 rewards the mean, not the median. Table I lists the chosen settings.

TABLE I
MAIN MODELS, TUNED GRIDS AND SETTINGS CHOSEN ON VALIDATION.

Model (target)	Grid $\rightarrow$ chosen
Ridge (returns)	$\alpha \in 0.1$–$10^9 \rightarrow 10^5$ (strong shrinkage: close to the zero forecast)
GB (hourly $\lvert r\rvert$)	learning rate $\{0.05, 0.1\}$, depth $\{2, 3, 6\} \rightarrow 0.1$, 6 (depth 3–6 within 0.004); 1,000 rows per leaf
GB (next-day vol.)	$\rightarrow$ learning rate 0.05, depth 2, 200 trees, 100 rows per leaf
Monthly refit	expanding, 24-month window, half-life 6/12 months $\rightarrow$ 12 months ($\lvert r\rvert$), expanding (daily)
LSTM ($\lvert r\rvert$)	2 layers, hidden $\{32, 64, 128\} \rightarrow 64$, dropout 0.2, Adam 10^{-3}, weight decay 10^{-4}, 5 seeds
MLP ($\lvert r\rvert$, long)	hidden (256, 128), weight decay 0.01
Timestral	28M/209M $\rightarrow$ 28M; series $\lvert r\rvert$; 180 days; clock, release calendar, range and volume, ETH size, VIX

F. Rationale for the Chosen Methods

Simple rules set the bar a model must beat. Regularised linear models suit a target that is mostly noise; trees capture non-linear effects of volatility and the clock; networks test whether a sequence model finds what hand-made features miss; the foundation model tests whether pretraining can replace fitting on our data. Monthly refits and conformal calibration answer the drift seen in the analysis.

III. Experimental Setup

A. Tools and Environment

All code is Python 3.12 with NumPy, pandas, SciPy, scikit-learn [20], PyTorch [21], the `timestral` package [19] and Matplotlib; versions are pinned in `requirements.txt`. The work consists of six Jupyter notebooks (base data, data analysis, classical models, neural networks, Timestral, trading simulation) sharing a `utils` package for the split dates, features, metrics and statistical tests. Code: https://github.com/Descalabreitor/IAI600-final-project.

B. Data Split

The data is split into three chronological blocks by the date of the predicted hours: *train* Jan 2020 – Dec 2022 (26,304

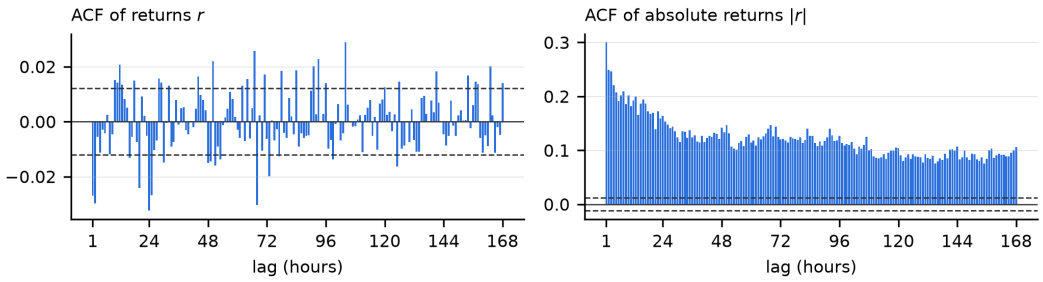


Fig. 1. Autocorrelation of hourly returns (left) and of their absolute value (right), lags 1–168 h, training block 2020–2022. Dashed lines: 95% band of white noise (too narrow under volatility clustering).

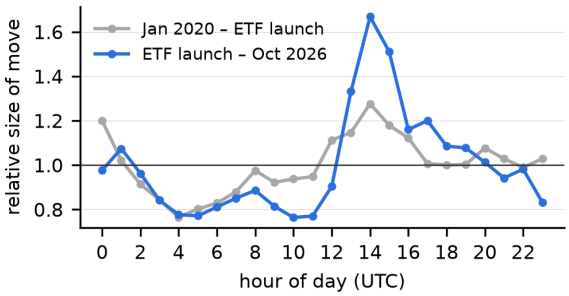


Fig. 2. Size of the hourly move by hour of day (UTC), relative to the average hour, before and after the launch of the US spot ETFs (11 Jan 2024). The 14 UTC peak already grows from 2022.

hours), *validation* Jan 2023 – Jun 2024 (13,128) and *test* Jul 2024 – Oct 2026 (19,899). A 192-hour purge before each border (the 168-hour LSTM window plus the 24 target hours) ensures that no target hour of one block lies in the next, so no model is fitted on hours it is later scored on [4]; inputs that reach further back are legitimate, since they are known at forecast time. Train and validation use every sixth origin; the test block uses every origin, and daily statistics use one forecast per day at 23:00 (829 independent test days).

C. Model Selection and Hyperparameter Tuning

Every choice (model, hyperparameters, features, refit schedule, calibration, Timestral setting) is made on train and validation only; the chosen model is refitted on both and scored once on test. Monthly refits and rolling calibration are walk-forward: during the test block they use an outcome only after it is known, as a forecaster in real time would. Hourly targets are tuned on the validation block, daily targets with four walk-forward folds over train and validation, which show the true spread between periods (± 0.08 R^2). Grids are small and extended when the best value lies on an edge. The number of boosting trees is tuned directly: early stopping on a random hold-out used up to 2.4 times too many. Models are selected by the reported metric, which matters for returns: a model trained for the mean but chosen by MAE is pushed towards the zero forecast. Networks stop early on the last six months of train, never on a random hold-out, and are averaged over five seeds.

D. Evaluation Metrics and Baselines

Returns are scored by the MAE skill against the zero forecast ($1 - \mathrm{MAE}/\mathrm{MAE}_0$), by R^2_{oos} (the R^2 against the zero forecast) and by the information coefficient (IC, the correlation of forecast and outcome). Direction is scored by the AUC and by the accuracy of the 10% most confident calls (binomial test); size and volatility by R^2; ranges by coverage, width and CRPS [22]. The R^2 of $|r|$ has a low ceiling: a single hour's $|r|$ is a very noisy measure of the volatility behind it, so even a perfect volatility forecast explains only part of its variance, and values of 0.1–0.2 already mean real skill. Baselines are the zero forecast, the reversal of the last hour, last week's volatility times the hour profile, and "always up". Uncertainty is measured by a moving-block bootstrap [23] over the test days (blocks of 10 days, 1,000 resamples) and Diebold–Mariano tests [24]; a difference is called *reliable* when its 95% interval excludes zero.

E. Trading Simulation

To see what the forecasts are worth, \$10,000 is traded spot (long or cash) with fixed rules: long only above the 50-day mean; a daily position $w = \min(1, 1.5\%/L)$, where L is the daily fall exceeded on 10% of days, $L = c\,\hat{\sigma}$ with c recalibrated on the last 90 known days (the rolling conformal idea); an hourly stop at $1.5L$; fees of 0.10% per side. Four flows differ only in the next-day risk $\hat{\sigma}$: last week's volatility, monthly GB, Timestral (summed from its 24 hourly forecasts) and the average of the two. The rules were fixed in advance, not tuned, so validation (a bull market) and test (mostly sideways) are two out-of-sample periods. We measure money (Sharpe, block bootstrap), the 10% Value-at-Risk with the Kupiec [25] and Christoffersen [26] backtests, drawdowns by block Monte Carlo (5,000 resampled years) and liquidations when futures leverage is set from each forecast.

IV. Results and Discussion

Table II summarises the test results; every model in it was chosen without the test block.

TABLE II
TEST-BLOCK RESULTS (JUL 2024 – OCT 2026) OF THE MODELS CHOSEN ON TRAIN AND VALIDATION, AGAINST THE BEST SIMPLE RULE. BRACKETS: 95% BLOCK-BOOTSTRAP INTERVALS.

Question	Chosen model	Test result	Simple rule	Beats rule?
24 hourly returns (assigned task)	tuned linear, KNN, trees, MLP, LSTM	MAE skill -0.0004 to -0.0002 (networks to -0.0020); IC ≤ 0.021; DM $p \geq 0.15$	zero forecast	no
Direction of the next hour	GB classifier (base + Binance)	AUC 0.538; 56.0% right on top-10% calls ($p = 5 \cdot 10^{-8}$)	last-hour reversal: AUC 0.528	yes, small
Size $\lvert r \rvert$ of each next hour	Timestral + GB refitted monthly	$R^2 = \mathbf{0.164}$ (Timestral 0.157, LSTM 0.156, GB 0.148)	last week $\times$ hour profile: 0.056	yes
80% range for each next hour	GB quantile + rolling Mondrian conformal	coverage 80.0% (calm 78%, wild 82%, release hours 82%), width 0.98%; Timestral uncalibrated: 80.3%, 1.00%	fixed range: 87.6%, 1.28%	yes
Next-day return	Ridge	$R^2_{\text{oos}} = -0.001$, IC 0.051	zero forecast	no
Next-day direction	Random Forest	AUC 0.509, accuracy 49.9%	always up: 50.1%	no
Next-day volatility	top-3 tree ensemble; GB refitted monthly	$R^2 = 0.510$ [0.45, 0.56]; 0.517	last week's mean: 0.094	yes
Trading \$10,000, risk-sized	GB + Timestral next-day risk	end \$12,278 (naive \$12,388); liquidations at $\approx 13\times$: 3 (naive 12); P(drawdown $> 20\%$) 20.8% (naive 27.2%)	last 7 days' volatility, same rules	safety only

A. Hourly Returns: the Assigned Task

No model forecasts the 24 hourly returns better than "no change". Every tuned classical model ends within 0.0001 percentage points of the MAE of the zero forecast (0.3193%), the long-format models have R^2_{oos} within ± 0.001 of zero (Diebold–Mariano $p \geq 0.15$), and the networks and the median of Timestral do no better. More model freedom always meant worse validation scores, and the chosen Ridge needs $\alpha = 10^5$, i.e. it shrinks almost to zero: the signature of a target that is mostly noise. Choosing models by the squared-error criterion they were trained for, rather than by MAE, raises the IC from 0.003 to 0.021, with an interval $[+0.006, +0.035]$ that excludes zero, but this ranking signal is too weak for the forecast itself to beat zero. At the hourly scale Bitcoin is close to a random walk, as Fig. 1 suggested and as market efficiency predicts [1]: a predictable direction would be traded away.

B. Next-Hour Direction

The one exception is the next hour. The rule "the next hour moves against the last one" reaches AUC 0.528 and is right 54.6% of the time on its 10% most confident calls ($p = 2 \cdot 10^{-5}$); GB with Binance order-flow features adds a little (AUC 0.538, 56.0%). An LSTM reaches the same AUC only when trained on every origin, which helps GB as much (0.541), so the data, not the network, makes the difference. The effect is real but small, measured before trading costs, and vanishes from the second hour on: hours 2–24 and the next day are a coin flip.

C. Size of the Hourly Moves

The size of the moves is where the models have clear skill (Fig. 3). The long format was the largest modelling gain (validation R^2 0.052 to 0.097, same model and features). On test, GB fitted once explains 13.1% of the variance of each hourly $\lvert r \rvert$, against 5.6% for the volatility rule (gain $[+0.057, +0.084]$). Refitting every month raises R^2 to 0.148 ($+0.017$ $[+0.010, +0.024]$, better in 20 of 29 months): in a drifting market (Fig. 2), keeping the model current is worth more than any extra input. None of the candidate features from the analysis, nor Binance and ETH, improved the trees (-0.0004 $[-0.004, +0.003]$ after refitting): recent volatility and the clock already carry their information. Release hours stay under-predicted (0.65 times their actual size): with 544 training rows they are fewer than one leaf of the tuned GB, and a correction factor learnt from past errors over-shot on test.

Networks gain only with care. An LSTM on the raw window reaches 0.100; telling it the clock and release flags of each predicted hour adds $+0.031$ (what the per-hour columns do for the trees), the volume attention $+0.014$ and monthly fine-tuning $+0.011$, all reliable, giving 0.156, narrowly above the refitted GB ($+0.008$ $[+0.001, +0.015]$). The MLP only ties the trees.

Timestral, used zero-shot, matches this without being trained on our data: 0.157 ($+0.009$ $[+0.002, +0.016]$ over refitted GB). What it reads matters more than its size: the series $\lvert r \rvert$ rather than r (validation 0.108 vs. 0.067), a long context (0.088, 0.105, 0.108 for 30, 90, 180 days) and the release calendar as known-future input ($+0.008$); the clock adds nothing, because the model finds the hour profile in its context. The 209M model added only 0.003 for seven times the compute. ETH size and VIX passed the validation rule by less than 0.001 and are part of the chosen setting, but add nothing on test. The plain average of Timestral and refitted GB, chosen on validation, is the best result of the project: $R^2 = 0.164$ ($+0.016$ $[+0.013, +0.020]$). Stacking Timestral's forecast as a GB feature was worse (0.155–0.157): the trees map it with what they learnt on the wilder 2020–2022 hours.

D. Prediction Ranges

The GB quantile ranges are 28% narrower than a fixed range (0.92% vs. 1.28%) and adapt to the market, but cover only 77.8% of the test hours: the test years differ from the training years. A correction measured once on validation does not

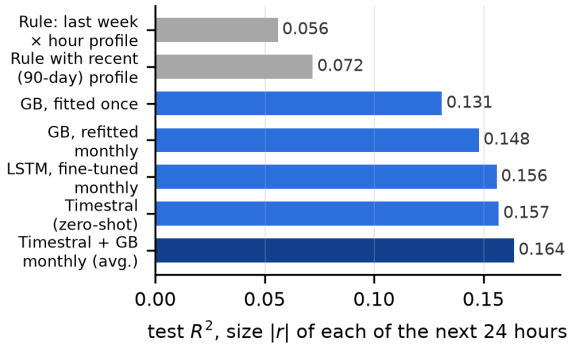


Fig. 3. Test R^2 for the size of each of the next 24 hourly moves: simple rules (grey), single models (blue) and the chosen average of Timestral and monthly-refitted Gradient Boosting (dark blue).

fix this (77.3%), whereas rolling conformal calibration, which keeps re-measuring the miss rate, does: 80.0% coverage, 78% in calm and 82% in wild hours, at 0.98% width. Release hours are the weak spot: one factor for all hours covers only 56% of them; the Mondrian variant, chosen on validation, lifts them to 82% with ranges 2.08% wide there. Timestral's quantiles need no calibration: 80.3% coverage at 1.00% width, every one of its 21 quantile levels within 1.4 points of its nominal level, and the lowest CRPS of the ranges tried (0.236 vs. 0.240 for a volatility-scaled baseline); thanks to the release calendar it covers 70% of the release hours without any calibration, below Mondrian but 13.8 points $[+7.9, +18.9]$ above a single rolling factor.

E. Next-Day Targets

At the daily scale, return and direction are not predictable (AUC interval $[-0.023, +0.046]$ around 0.5), but volatility is: a top-3 tree ensemble, chosen on walk-forward folds, explains 51% of the next day's log volatility, against 9.4% for last week's average. A monthly refit of GB raises its R^2 from 0.503 to 0.517 ($+0.015$ $[+0.004, +0.025]$), and averaging GB with an MLP adds $+0.011$ $[+0.001, +0.022]$ over GB. On a small daily table of the next day's move size, plain Linear Regression ($R^2 = 0.514$) and SVR (0.519) beat Random Forest (0.464), because tomorrow's volatility depends almost linearly on today's.

F. Trading Simulation

A first economic test, volatility targeting, lost to buy-and-hold (Sharpe 0.27 vs. 0.50, not significant) because of its design (target level, leverage, turnover); the trading simulation isolates the forecast instead. As risk meters the models are much better than last week's volatility (test R^2 of next-day log volatility, bias removed: 0.54 GB, 0.55 Timestral, 0.57 their average, 0.17 naive), but they do not earn more money: on test the four flows end at \$11,813–12,388, the naive one highest; every Sharpe difference to it has an interval including zero; and a model flow ends ahead in at most 3 of 12 other budget and stop settings. A risk-sized long position can only be smaller or larger, never on the right side: in the bull market of validation, cutting it before volatile days cost money, because those days were mostly up days.

The forecasts do make trading safer. Setting the leverage daily from the averaged forecast, at the same average leverage as the naive flow ($\approx 13\times$ on test), leads to 3 liquidations instead of 12; the probability of a drawdown above 20% within a year falls from 27.2% to 20.8%; and only the average passes both VaR backtests in both periods (Timestral's breaches cluster on test, Christoffersen $p = 0.01$). An hourly stop that stays tight until the forecast wild hours have passed, plus a 7-day loss limit, lowers that probability to 10–12%, against 54% for the trend rule without risk control (buy-and-hold: maximum drawdown 53%); most of this comes from the rules themselves (11.9% with naive estimates). Given the signed returns, Timestral does not predict the direction either (daily AUC 0.48–0.51), and trading on its expected return lost money on test.

G. Discussion

Because Bitcoin's behaviour drifts, models and calibrations that are kept current beat models with more data or more inputs. A feature that carries information on its own does not help a model that already sees recent volatility and the clock, or that has too few examples to use it. The trading simulation repeats the central result in money terms: a size forecast without a direction cannot earn more, but it can control losses. Model families converge: GB, a carefully built LSTM and an untrained foundation model end within 0.01 R^2, and averaging two different families gives the best result. Two caveats apply: there is a single test period of 2.25 years, and Timestral was released after our test block without a list of its training series; if Bitcoin 2023–2026 is among them, its scores are optimistic.

V. Conclusion

The 24 hourly price changes of Bitcoin's next day cannot be forecast better than "no change": at the hourly scale the price is close to a random walk, and only the next hour carries a small reversal signal (AUC 0.54). The size of the changes is predictable, so we recommend answering the task with the expected size and an 80% range for each hour. The average of zero-shot Timestral and monthly-refitted Gradient Boosting explains 16.4% of the hourly variance (2.9 times a volatility rule), and conformally calibrated ranges cover 80% of the hours, release hours included. Used to size trades, these forecasts do not earn more money than a naive volatility estimate, but they make trading safer: far fewer liquidations at the same leverage and a lower chance of large drawdowns.

We learned that the evaluation design matters more than the choice of model: chronological blocks with a purge, choices made without the test block, comparison with simple rules and bootstrap intervals are what separate a real signal from noise. Recency matters more than data volume or extra sources: monthly refits were the largest single gain, while the new features, ETH and macro variables added nothing reliable to the size forecasts, and Binance order flow only a little to the

next-hour direction. A pretrained model, used as it comes, can match careful feature engineering where there is signal, but cannot create signal where there is none. And a good forecast is not a strategy: knowing the size of the moves without their direction controls losses, not returns.

The study has limitations: a single test period; many tests on the return target without a multiple-testing correction; no trading costs in the next-hour signal; release hours that remain hard to forecast; two trading periods with no significance test of the safety gains and no slippage or funding costs; and a possible overlap between Timestral's training data and our test period. Future work includes a rolling-origin evaluation over 2023–2026, Holm or Benjamini–Hochberg corrections [8], a cost-aware test of the next-hour signal, a forward test of Timestral on data after its release, and options, where the size of a move pays without its direction.

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Contributions. Alberto Prats: classical models and hyperparameter tuning; Adrián Perera: data pipeline, unification of the notebooks, neural networks and Timestral; Filip Botlo: Binance features, monthly refits, conformal ranges, course-method pipeline, trading simulation; Adrián Muñiz: exploratory data analysis and macro correlations. We thank the author of the Kaggle dataset, Binance, FRED and Mistral AI for the public data and models, and the course staff.

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